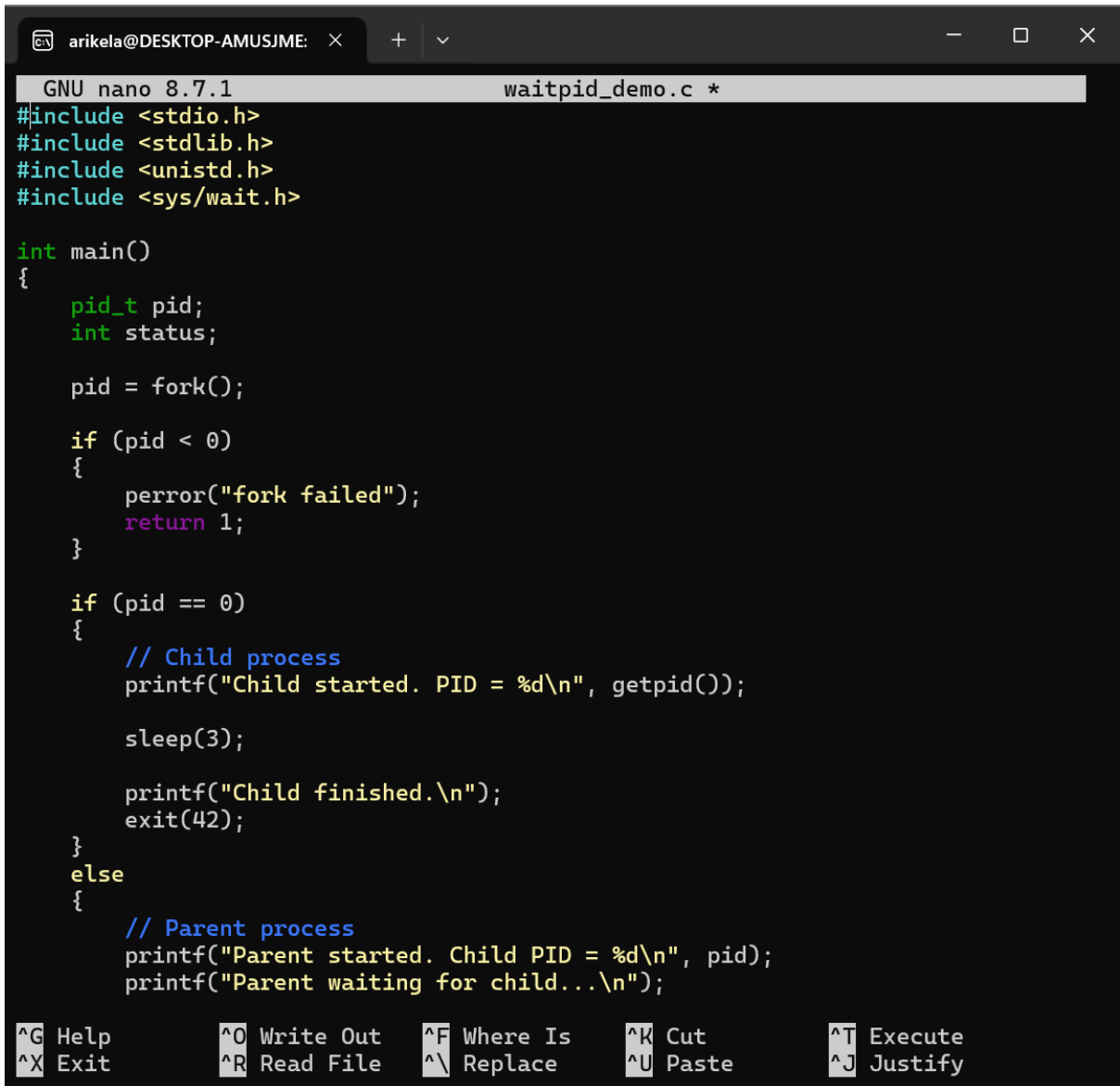


Skill Week 7

Task-1: Process Synchronization and Child Process Monitoring Using waitpid()



```
GNU nano 8.7.1 waitpid_demo.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    int status;

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        // Child process
        printf("Child started. PID = %d\n", getpid());

        sleep(3);

        printf("Child finished.\n");
        exit(42);
    }
    else
    {
        // Parent process
        printf("Parent started. Child PID = %d\n", pid);
        printf("Parent waiting for child...\n");
    }
}
```

Help **Write Out** **Where Is** **Cut** **Execute**
Exit **Read File** **Replace** **Paste** **Justify**

```

// Parent process
printf("Parent started. Child PID = %d\n", pid);
printf("Parent waiting for child...\n");

waitpid(pid, &status, 0);

if (WIFEXITED(status))
{
    printf("Child terminated normally.\n");
    printf("Child exit status = %d\n", WEXITSTATUS(status));
}

return 0;
}
|

```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify

```

arikela@DESKTOP-AMUSJME:~$ cd skillweek7
arikela@DESKTOP-AMUSJME:~/skillweek7$ nano waitpid_demo.c
arikela@DESKTOP-AMUSJME:~/skillweek7$ gcc waitpid_demo.c -o waitpid_demo
arikela@DESKTOP-AMUSJME:~/skillweek7$ ./waitpid_demo
Child started. PID = 1086
Parent started. Child PID = 1086
Parent waiting for child...
Child finished.
Child terminated normally.
Child exit status = 42
arikela@DESKTOP-AMUSJME:~/skillweek7$

```

Task-2: To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution

arikela@DESKTOP-AMUSJME: × + ▾

GNU nano 8.7.1path_resolver.c *

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/stat.h>

#define MAX_PATH 4096

int main()
{
    char command[100];
    char *path;
    char *path_copy;
    char *directory;

    printf("Enter command: ");
    scanf("%99s", command);

    // Retrieve PATH
    path = getenv("PATH");

    if (path == NULL)
    {
        printf("PATH variable not found.\n");
        return 1;
    }

    // Make a copy because strtok modifies the string
    path_copy = malloc(strlen(path) + 1);

    if (path_copy == NULL)
    {
        perror("malloc");
    }
}
```

^G Help
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 ^R Read File

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 ^\ Replace

^K Cut
 ^U Paste

^T Execute
 ^J Justify

```
GNU nano 8.7.1 path_resolver.c *
// Make a copy because strtok modifies the string
path_copy = malloc(strlen(path) + 1);

if (path_copy == NULL)
{
    perror("malloc");
    return 1;
}

strcpy(path_copy, path);

// Parse PATH directories
directory = strtok(path_copy, ":");

while (directory != NULL)
{
    char full_path[MAX_PATH];

    snprintf(full_path, sizeof(full_path),
             "%s/%s", directory, command);

    // Check whether executable exists
    if (access(full_path, X_OK) == 0)
    {
        printf("Executable found: %s\n", full_path);

        free(path_copy);
        return 0;
    }

    directory = strtok(NULL, ":");
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
printf("Command not found: %s\n", command);

free(path_copy);

return 1;
}
```

```
arikela@DESKTOP-AMUSJME:~/skillweek7$ nano path_resolver.c
arikela@DESKTOP-AMUSJME:~/skillweek7$ gcc path_resolver.c -o path_resolver
arikela@DESKTOP-AMUSJME:~/skillweek7$ ./path_resolver
Enter command: ls
Executable found: /usr/bin/ls
arikela@DESKTOP-AMUSJME:~/skillweek7$ ./path_resolver
Enter command: gcc
Executable found: /usr/bin/gcc
arikela@DESKTOP-AMUSJME:~/skillweek7$ ./path_resolver
Enter command: hello123
Command not found: hello123
arikela@DESKTOP-AMUSJME:~/skillweek7$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporatio
nII.WindowsSubsystemForLinux_2.7.10.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/C
ommon Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle
/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/m
nt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Window
s/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Prog
ram Files/dotnet:/mnt/c/MinGW/bin:/mnt/c/Program Files/Java/jdk-25/bin:/mnt/c/
Program Files/nodejs:/mnt/c/ProgramData/chocolatey/bin:/mnt/c/Program Files/Gi
t/cmd:/mnt/c/Users/arika/.local/bin:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/
bin:/mnt/c/Users/arika/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/arika/
AppData/Local/Programs/Microsoft VS Code/bin:/mnt/d/cprog/S10C/bin:/mnt/c/Users
/arika/AppData/Local/GitHubDesktop/bin:/mnt/c/Users/arika/AppData/Roaming/npm:/
mnt/c/Users/arika/AppData/Local/Python/bin:/snap/bin
```